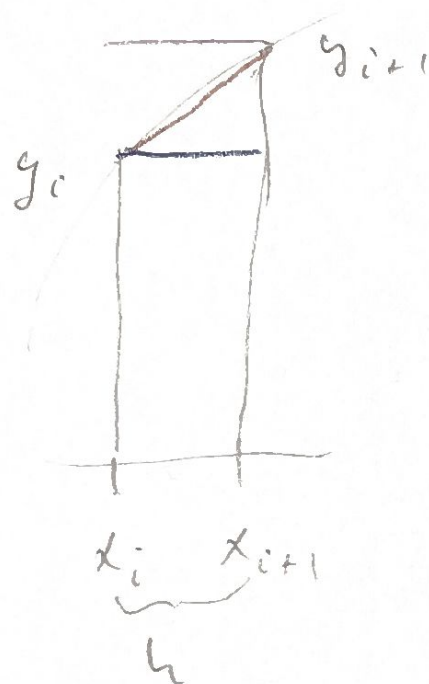
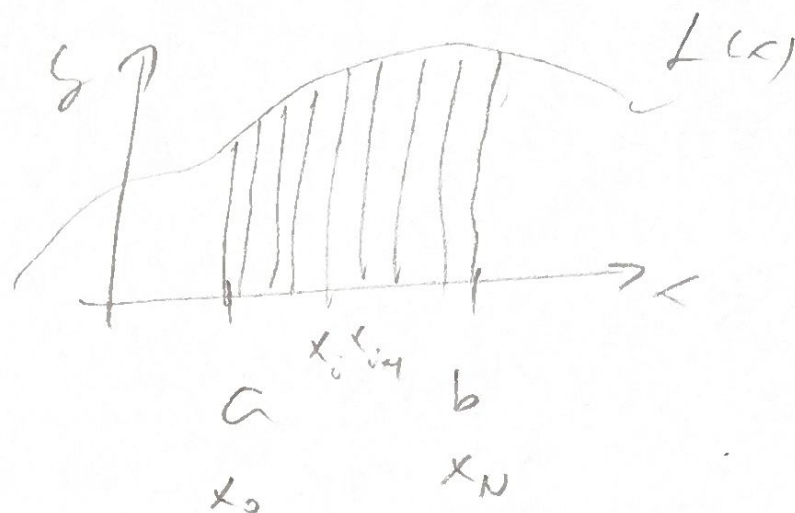
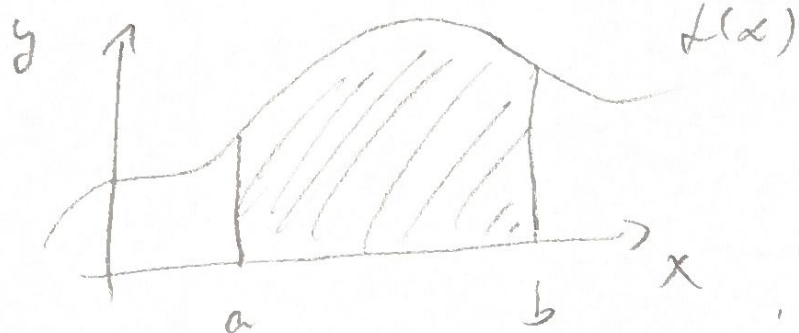


Introduction to Numerical Methods - 6 -

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Integration:



$$I_i^{(0)} = y_i \cdot h$$

$$I_i^{(R)} = y_{i+1} \cdot h$$

$$I_i^{(K)} = \frac{y_i + y_{i+1}}{2} \cdot h$$

$$I^{(L)} = \sum_{i=0}^{N-1} x_i \cdot h$$

$$I^{(R)} = \sum_{i=1}^N x_i \cdot h$$

$$I^{(K)} = \sum_{i=0}^{N-1} \frac{x_i + x_{i+1}}{2} \cdot h = \sum_{i=0}^{N-1} \frac{x_i}{2} \cdot h + \sum_{i=0}^{N-1} \frac{x_{i+1}}{2} \cdot h = \frac{I^{(L)} + I^{(R)}}{2}$$

$$= \frac{x_0 + x_N}{2} + \sum_{i=1}^{N-1} x_i \cdot h$$

Only the boundary is different

Riemann error:

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Taylor in the point x_i

$$f(x) \approx f(x_i) + f'(x_i)(x - x_i) + \frac{1}{2}f''(x_i)(x - x_i)^2 + \dots$$

$$\int_{x_i}^{x_{i+1}} f(x) dx = \int_{x_i}^{x_{i+1}} dx \left(f(x_i) + f'(x_i)(x - x_i) + \dots \right)$$

$$\int_{x_i}^{x_{i+1}} f(x) dx = h \cdot f(x_i) + \frac{h^2}{2} f'(x_i) + \mathcal{O}(h^3)$$

$$\int_{x_i}^{x_{i+1}} f(x) dx = h f(x_i) + \mathcal{O}(h^2)$$

We have to do this N times

$$N = \frac{b-a}{h} \Rightarrow \underline{\text{The total error is } \frac{b-a}{h} \mathcal{O}(h^2) = \mathcal{O}(h)}$$

Trapezoid error:

$$\begin{cases} f(x) = f(x_i) + f'(x_i)(x - x_i) + \frac{1}{2}f''(x_i)(x - x_i)^2 + \dots \\ f(x) = f(x_{i+1}) + f'(x_{i+1})(x_{i+1} - x) + \frac{1}{2}f''(x_{i+1})(x_{i+1} - x)^2 + \dots \end{cases}$$

$$y_i = \frac{x_i + x_{i+1}}{2}$$

Taylor for x_i, x_{i+1} around y_i

$$f(x_i) = f(y_i) + f'(y_i)(x_i - y_i) + f''(y_i) \underbrace{\left(\frac{x_i - y_i}{2} \right)^2}_{h^2/4} + \dots$$

$$f(x_{i+1}) = f(y_i) + f'(y_i) \underbrace{(x_{i+1} - y_i)}_{+h/2} + f''(y_i) \underbrace{\left(\frac{x_{i+1} - y_i}{2} \right)^2}_{h^2/4} + \dots$$

$$\frac{f(x_i) + f(x_{i+1})}{2} = f(y_i) + \mathcal{O}(h^2)$$

$$\int_{x_i}^{x_{i+1}} f(x) dx = \int_{x_i}^{x_{i+1}} \left(f(y_i) + f'(y_i)(x - y_i) + \dots \right) dx =$$

↖ this disappears
odd function!

$$= hf(y_i) + O(h^3)$$

Total Error: Trapezoidal: $O(h^2)$

Higher order methods:

e.g. Simpson's Rule: fit a parabola to three points
lower error, but difference is only at
the boundary

It turns out that it is important
where we take the sample points.
Select them wisely!

The general formula, the Gauss-Legendre
quadrature is the following:

$$\int_{-1}^1 f(x) dx \approx \sum_{i=0}^{n-1} w_i f(x_i)$$

- n : number of points used
- x_i : sampling points
- w_i : weights

Example: $n=1$ to be determined w_0, x_0

It can exactly approximate polynomials of degree 1 because they have two parameters

$$f(x) = Ax + B$$

$$\int_{-1}^1 f(x) dx = w_0 \cdot f(x_0)$$

$$\int_{-1}^1 (Ax + B) dx = \left[\frac{A}{2} x^2 + Bx \right]_{-1}^1 = 2B = w_0(Ax_0 + B) = w_0Ax_0 + Bw_0$$

$$\Rightarrow x_0 = 0, w_0 = 2, \text{ so } I = 2 \cdot f(0)$$

Which is just the center point integral.

$n=2$:

4 unknowns: $w_0, w_1, x_0, x_1 \Rightarrow$ 3rd degree polyn.

$$\int_{-1}^1 f(x) dx = w_0 f(x_0) + w_1 f(x_1)$$

$$\int_{-1}^1 (Ax^3 + Bx^2 + Cx + D) dx = \frac{2}{3}B + 2D =$$

$$= w_0(Ax_0^3 + Bx_0^2 + Cx_0 + D) + w_1(Ax_1^3 + Bx_1^2 + Cx_1 + D)$$

Equations:

$$w_0 x_0^3 + w_1 x_1^3 = 0$$

$$w_0 x_0^2 + w_1 x_1^2 = \frac{2}{3}$$

$$w_0 x_0 + w_1 x_1 = 0$$

$$w_0 + w_1 = 2$$

} solution

$$w_0 = w_1 = 1$$

$$x_0 = -\frac{1}{\sqrt{3}}, x_1 = \frac{1}{\sqrt{3}} \approx \pm 0.577$$

closer to the boundary

General case:

x_i : are the roots of the n th Legendre polynomial

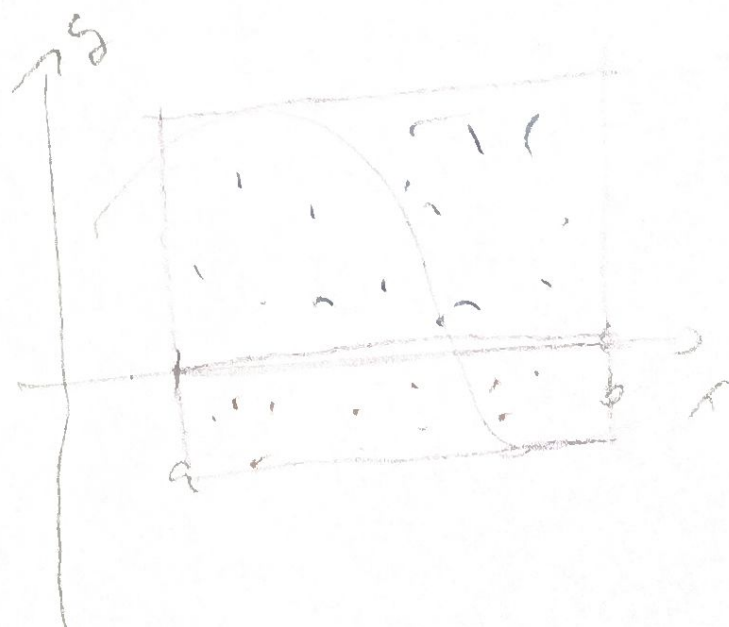
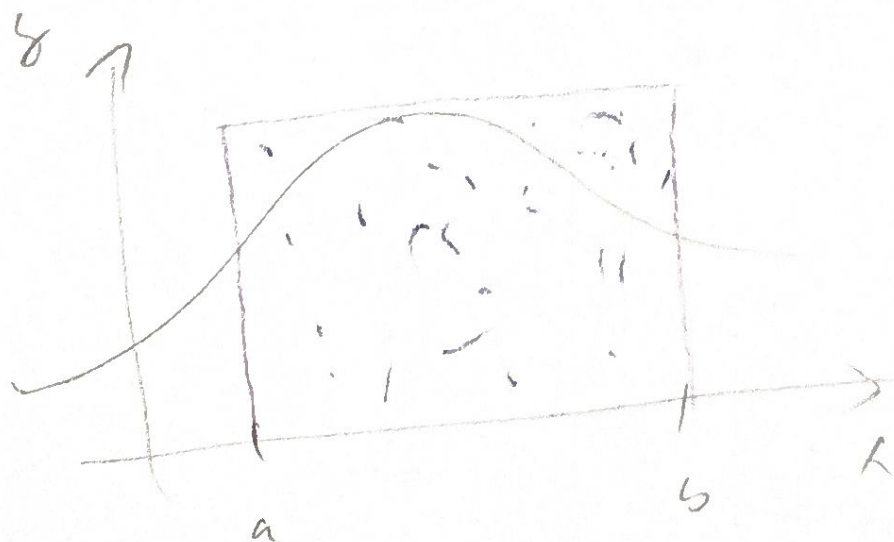
w_i : are linear in the equations

n	x_i	w_i
1	0	2
2	$\pm \frac{1}{\sqrt{3}}$	1
3	0, $\pm \frac{\sqrt{3}}{2}$	$\frac{8}{9}, \frac{5}{9}$
4	$\pm \sqrt{\frac{3}{7} - \frac{2}{7}\sqrt{\frac{6}{5}}}$	$\frac{18 + \sqrt{30}}{36}$
	$\pm \sqrt{\frac{3}{7} + \frac{2}{7}\sqrt{\frac{6}{5}}}$	$\frac{18 - \sqrt{30}}{36}$

Monte-Carlo integral

When everything fails!

Box the integral in the $x-y$ plane.



- Generate random points in the box.
- Count the fraction of points under the $f(x)$ function.
- Calculate the area of the box A
- The integral is $I = f \cdot A$

(If $f(x)$ has negative values in $[a, b]$ then do it separately for $y > 0$, and $y < 0$)